

LECTURE 10

The Governing Equations: Euler and Navier–Stokes

Reference: J. C. Tannehill, D. A. Anderson & R. H. Pletcher, Computational Fluid Mechanics and Heat Transfer, 2nd ed., Taylor & Francis, 1997. Chapter 5, Section 5.1.

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Euler and Navier Stokes equations

Three conservation laws:

- Mass
- Momentum
- Energy

each written for a small element of fluid.

Different physics:

- Inviscid gives the Euler equations
- Viscous and heat-conducting gives Navier–Stokes equations.

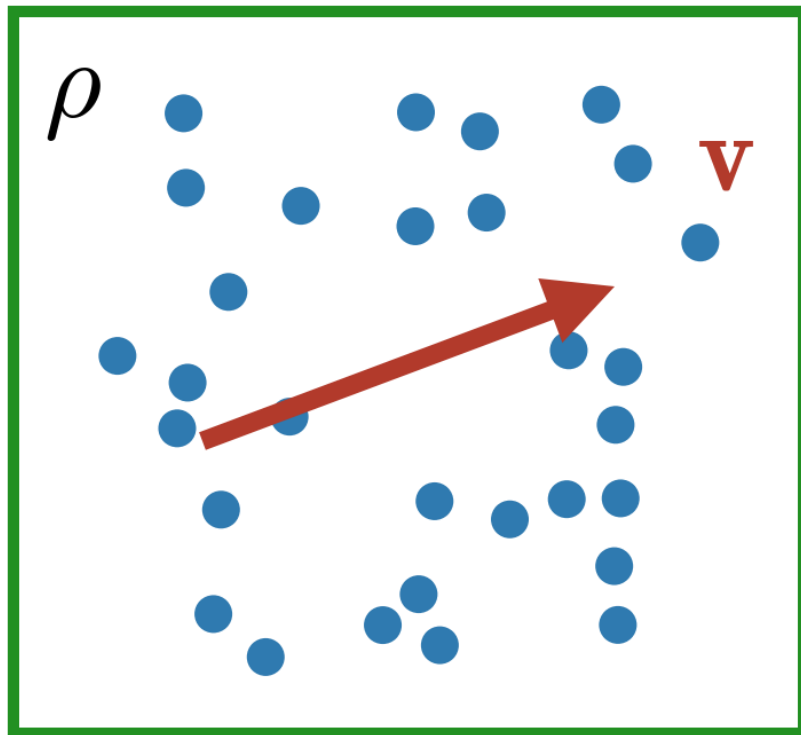
Two regimes:

Compressible: where p is an unknown.

Incompressible: where it is not.

The fluid as a continuum

A fluid element is large enough to contain many particles / molecules and small compared with the scale of the flow.



many molecules in a volume
small compared with the flow

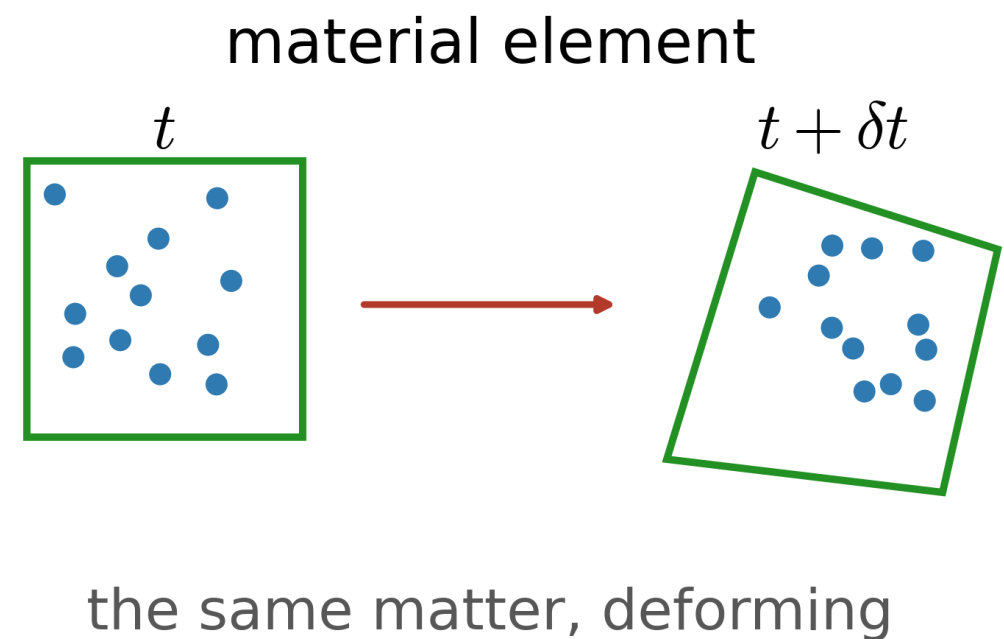
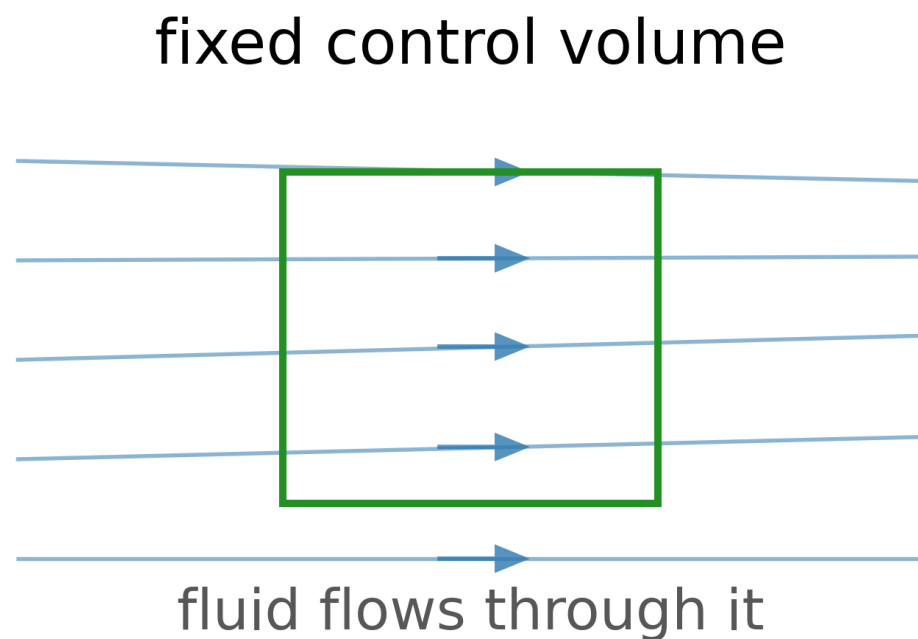
$$\rho = \frac{\delta m}{\delta \mathcal{V}}$$

a fluid element

Density, velocity, pressure and temperature are then smooth functions of position and time, and we may differentiate them.

$$\mathbf{v} = u \hat{i} + v \hat{j} + w \hat{k}, \quad \rho = \rho(x, y, z, t), \quad p = p(x, y, z, t)$$

Two fluid views



Fixed control volume: a region fixed in space, with fluid flowing through it. Gives the conservation form directly.

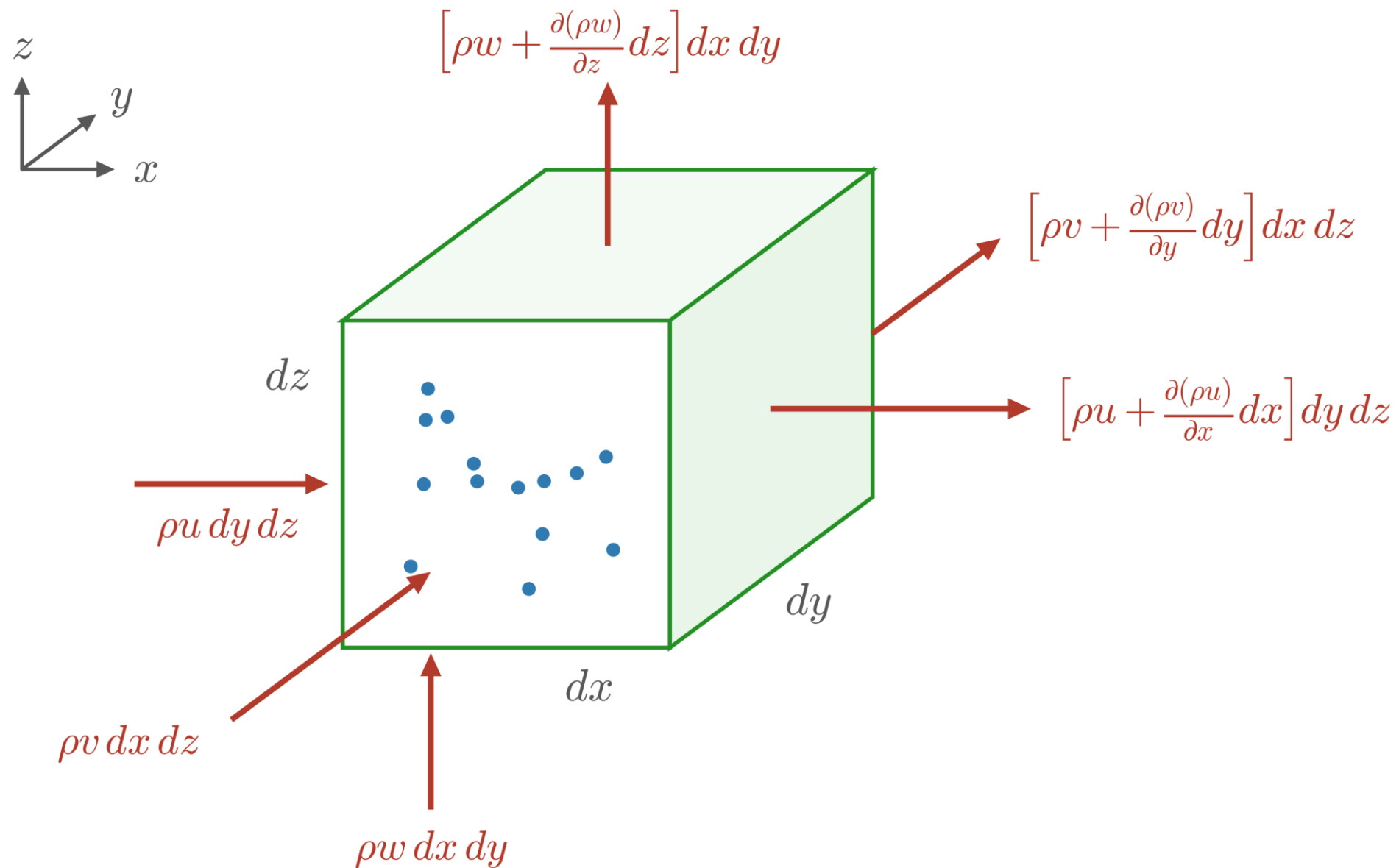
Material element: the same piece of matter, followed as it moves and deforms. Gives Newton's second law directly.

The 2 are connected by the material derivative, which we studied earlier:

$$\frac{D(\)}{Dt} = \frac{\partial(\)}{\partial t} + (\mathbf{v} \cdot \nabla)(\) = \frac{\partial(\)}{\partial t} + u \frac{\partial(\)}{\partial x} + v \frac{\partial(\)}{\partial y} + w \frac{\partial(\)}{\partial z}$$

Conservation of mass

Take a fixed box of sides dx , dy , dz . Mass is neither created nor destroyed, so what accumulates is what enters minus what leaves.



Mass flux through the faces of a fixed element. The flux per unit area is ρ times the velocity component normal to the face.

Mass: the net flux

The net mass leaving through the two x faces per unit time is the difference between them:

$$\left[\rho u + \frac{\partial(\rho u)}{\partial x} dx \right] dy dz - \rho u dy dz = \frac{\partial(\rho u)}{\partial x} dx dy dz$$

The same holds for y and z .

The mass inside the box changes at the rate $(\partial\rho/\partial t) dx dy dz$, so:

$$\frac{\partial\rho}{\partial t} dx dy dz + \left[\frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} + \frac{\partial(\rho w)}{\partial z} \right] dx dy dz = 0$$

The continuity equation

Dividing by $dx\,dy\,dz$ and taking the limit as the box shrinks to a point:

$$\frac{\partial \rho}{\partial t} + \frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} + \frac{\partial(\rho w)}{\partial z} = 0$$

The 3 flux terms are the divergence of $\rho \mathbf{v}$, so in vector form:

$$\boxed{\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0}$$

This is the conservation form:

A time derivative plus the divergence of a flux.

Every conservation law has this shape.

Other forms of the continuity equation

Expanding the divergence with the product rule and collecting the material derivative:

$$\nabla \cdot (\rho \mathbf{v}) = \mathbf{v} \cdot \nabla \rho + \rho \nabla \cdot \mathbf{v} \quad \implies \quad \frac{D\rho}{Dt} + \rho \nabla \cdot \mathbf{v} = 0$$

This is the non-conservative form:

The density of a particle changes only if the flow diverges.

If the fluid is incompressible, the density of each particle is constant, and the equation becomes a constraint on the velocity:

$$\frac{D\rho}{Dt} = 0 \quad \implies \quad \nabla \cdot \mathbf{v} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$$

Conservation of momentum

Newton's second law applies to matter, so we take a material element and divide by its volume:

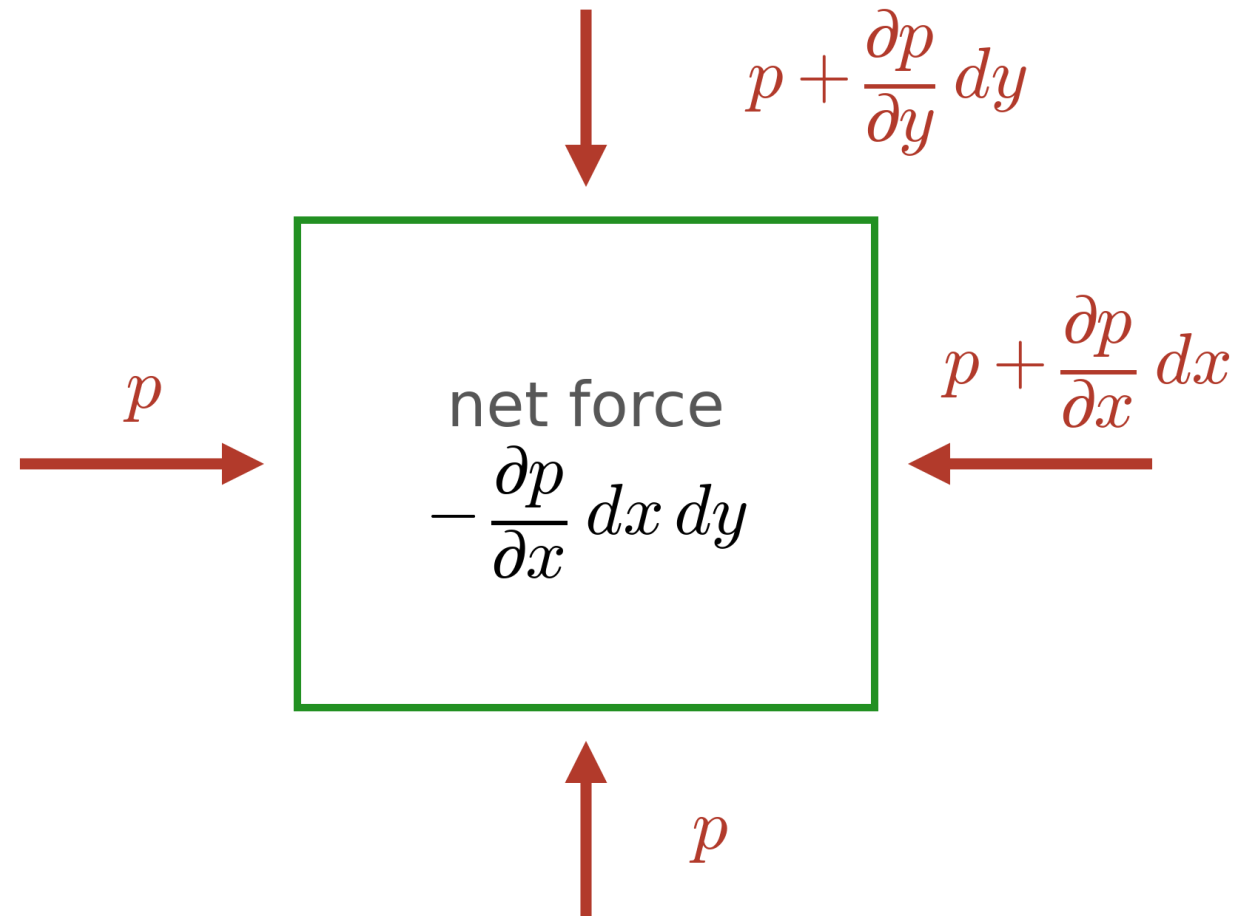
$$\delta m \frac{D\mathbf{v}}{Dt} = \delta \mathbf{F} \quad \implies \quad \rho \frac{D\mathbf{v}}{Dt} = \mathbf{f}$$

Two kinds of force act on it:

- **Body forces** act on the mass throughout the volume (we drop them initially).
- **Surface forces** act on the faces, transmitted by the fluid outside (we consider them).

The pressure force

In a fluid at rest the only surface force is pressure, acting normal to each face and pushing inwards.



The two x faces do not cancel, because the pressure on the right is larger by $(\partial p / \partial x) dx$:

$$p dy dz - \left[p + \frac{\partial p}{\partial x} dx \right] dy dz = - \frac{\partial p}{\partial x} dx dy dz$$
$$\implies \mathbf{f} = -\nabla p$$

The Euler momentum equation

With pressure as the only surface force, Newton's law for a fluid element becomes:

$$\rho \frac{D\mathbf{v}}{Dt} = -\nabla p$$

Written out, three scalar equations, each one an equation for a velocity component:

$$\rho \left(\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} \right) = -\frac{\partial p}{\partial x}$$

$$\rho \left(\frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} \right) = -\frac{\partial p}{\partial y}$$

$$\rho \left(\frac{\partial w}{\partial t} + u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} + w \frac{\partial w}{\partial z} \right) = -\frac{\partial p}{\partial z}$$

Momentum in conservation form

The material form is not what a finite-volume code considers:

$$\rho \frac{D\mathbf{v}}{Dt} = -\nabla p$$

We expand the conservation form of the x momentum and use continuity:

$$\frac{\partial(\rho u)}{\partial t} + \nabla \cdot (\rho u \mathbf{v}) = \rho \frac{Du}{Dt} + u \underbrace{\left[\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) \right]}_{=0}$$

The bracket is the continuity equation, so it vanishes. The two forms are therefore identical:

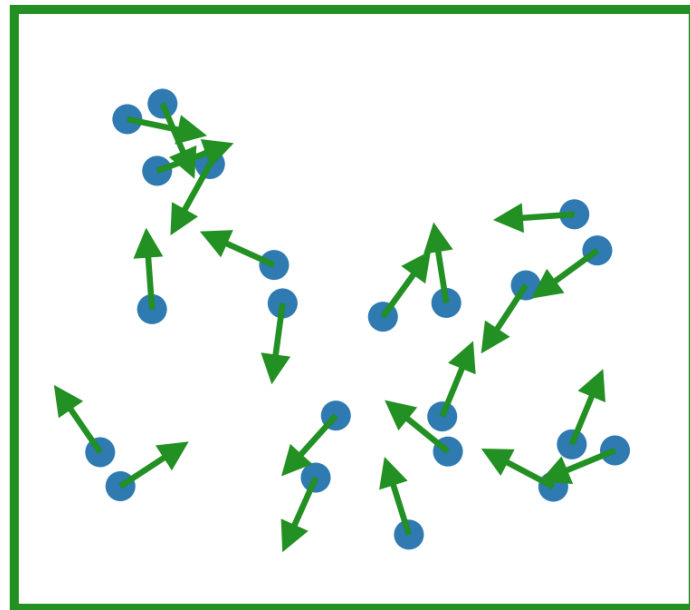
$$\frac{\partial(\rho \mathbf{v})}{\partial t} + \nabla \cdot (\rho \mathbf{v} \mathbf{v} + p \mathbf{I}) = 0$$

Conservation of energy

The first law of thermodynamics for a fluid element:

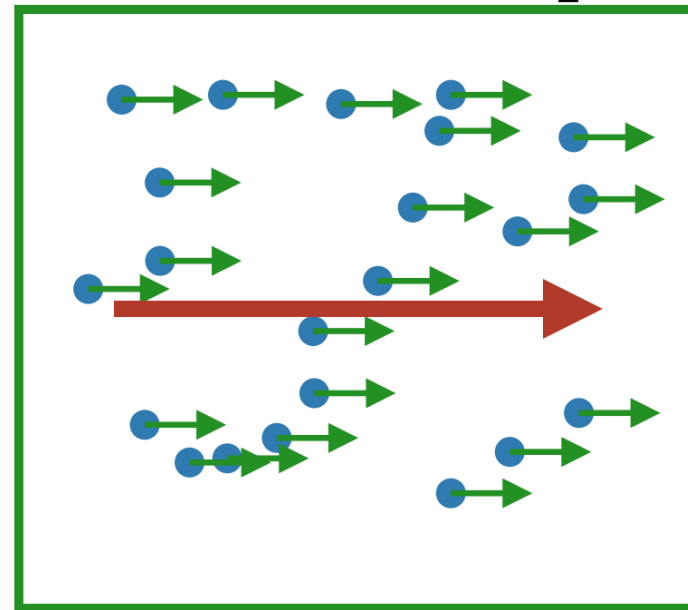
the energy inside changes by the heat added and the work done on it.

internal energy ρe



random motion

kinetic energy $\frac{1}{2}\rho |\mathbf{v}|^2$



bulk motion

The energy per unit volume has 2 parts: the random motion of the particles/molecules and the organised motion of the element:

$$E_t = \rho \left(e + \frac{1}{2} |\mathbf{v}|^2 \right) = \rho e + \frac{1}{2} \rho (u^2 + v^2 + w^2)$$

The inviscid energy equation

With no conduction and no viscosity, the only work done on the element is by pressure:

$$\dot{W} = -\nabla \cdot (p \mathbf{v}), \quad \dot{Q} = 0, \quad \Phi = 0$$

The first law per unit volume then reads:

$$\frac{\partial E_t}{\partial t} + \nabla \cdot (E_t \mathbf{v}) = -\nabla \cdot (p \mathbf{v})$$

Moving the pressure work inside the divergence gives the conservation form:

$$\boxed{\frac{\partial E_t}{\partial t} + \nabla \cdot [(E_t + p) \mathbf{v}] = 0}$$

The flux carries the **total enthalpy**, not the energy alone.

The enthalpy form of the flux

The combination $E_t + p$ is related to the enthalpy:

$$h = e + \frac{p}{\rho}, \quad h_0 = h + \frac{1}{2}|\mathbf{v}|^2$$

Where h is the static enthalpy and h_0 is the total enthalpy including the kinetic energy. Then:

$$E_t + p = \rho \left(e + \frac{p}{\rho} + \frac{1}{2}|\mathbf{v}|^2 \right) = \rho \left(h + \frac{1}{2}|\mathbf{v}|^2 \right) = \rho h_0$$

Thus, the flux is the total enthalpy carried by the flow:

$$\boxed{\frac{\partial E_t}{\partial t} + \nabla \cdot (\rho h_0 \mathbf{v}) = 0}$$

In steady inviscid flow h_0 is constant along a streamline.

The compressible Euler equations

All three laws have the same shape, so they combine into one equation:

$$\frac{\partial \Psi}{\partial t} + \frac{\partial \Theta_x}{\partial x} + \frac{\partial \Theta_y}{\partial y} + \frac{\partial \Theta_z}{\partial z} = 0$$

$$\Psi = \begin{bmatrix} \rho \\ \rho u \\ \rho v \\ \rho w \\ E_t \end{bmatrix}, \quad \Theta_x = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ \rho uw \\ (E_t + p)u \end{bmatrix}$$

$$\Theta_y = \begin{bmatrix} \rho v \\ \rho uv \\ \rho v^2 + p \\ \rho vw \\ (E_t + p)v \end{bmatrix}, \quad \Theta_z = \begin{bmatrix} \rho w \\ \rho uw \\ \rho vw \\ \rho w^2 + p \\ (E_t + p)w \end{bmatrix}$$

Ψ holds the conserved variables; Θ with a direction subscript is the flux in that direction.

We have 5 equations with 5 conserved variables.

There are no second derivatives anywhere.

The incompressible Euler equations

Set the density constant and keep the fluid inviscid. Continuity becomes a constraint:

$$\nabla \cdot \mathbf{v} = 0$$

$$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} = -\frac{1}{\rho} \nabla p$$

Written out, 3 momentum equations:

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} = -\frac{1}{\rho} \frac{\partial p}{\partial x}$$

$$\frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} = -\frac{1}{\rho} \frac{\partial p}{\partial y}$$

$$\frac{\partial w}{\partial t} + u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} + w \frac{\partial w}{\partial z} = -\frac{1}{\rho} \frac{\partial p}{\partial z}$$

4 equations for u , v , w and p . The energy equation decouples: with constant density T no longer feeds back.

Closing the system

We counted 5 conservation equations, and more unknowns than that:

5 conservation equations + 2 state relations

7 unknowns: ρ, u, v, w, e, p, T

Two thermodynamic relations close it. For a calorically perfect gas:

$$\boxed{\begin{aligned} p &= \rho R T, & e &= c_v T, & \Gamma &= \frac{c_p}{c_v} \\ \implies & & p &= (\Gamma - 1) \rho e \end{aligned}}$$

Nothing more is needed for inviscid flow.

The speed of sound follows:

$$c_s \equiv a = \sqrt{\Gamma \frac{p}{\rho}} = \sqrt{\Gamma R T}, \quad \text{M} = \frac{|\mathbf{v}|}{c_s}$$

Some codes fix Γ , so $p = (\Gamma - 1)\rho e$ already contains both.

Closing the incompressible system

The density is given, so no equation of state is needed. Count again:

4 equations + $\rho = \text{constant}$

4 unknowns: u, v, w, p

But no equation evolves the pressure. Take the divergence of the momentum equation and use the constraint:

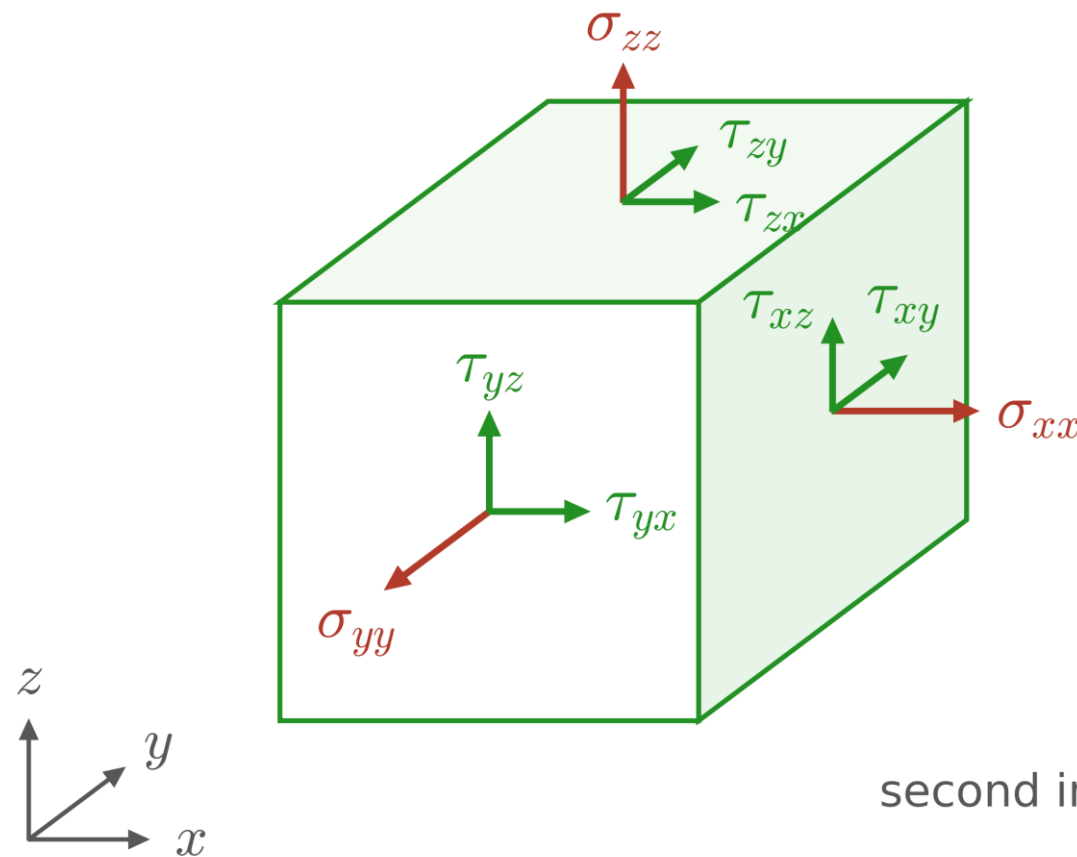
$$\nabla^2 p = -\rho \frac{\partial u_i}{\partial x_j} \frac{\partial u_j}{\partial x_i}$$

The pressure is not a thermodynamic variable here. It is whatever keeps the flow divergence free.

An elliptic equation is solved at every step: incompressibility.

Real fluids: the stress tensor

A moving fluid also transmits tangential force across a surface. Nine components are needed: three faces, three directions each.



first index: the face
second index: the push direction
red: normal stress
green: shear stress

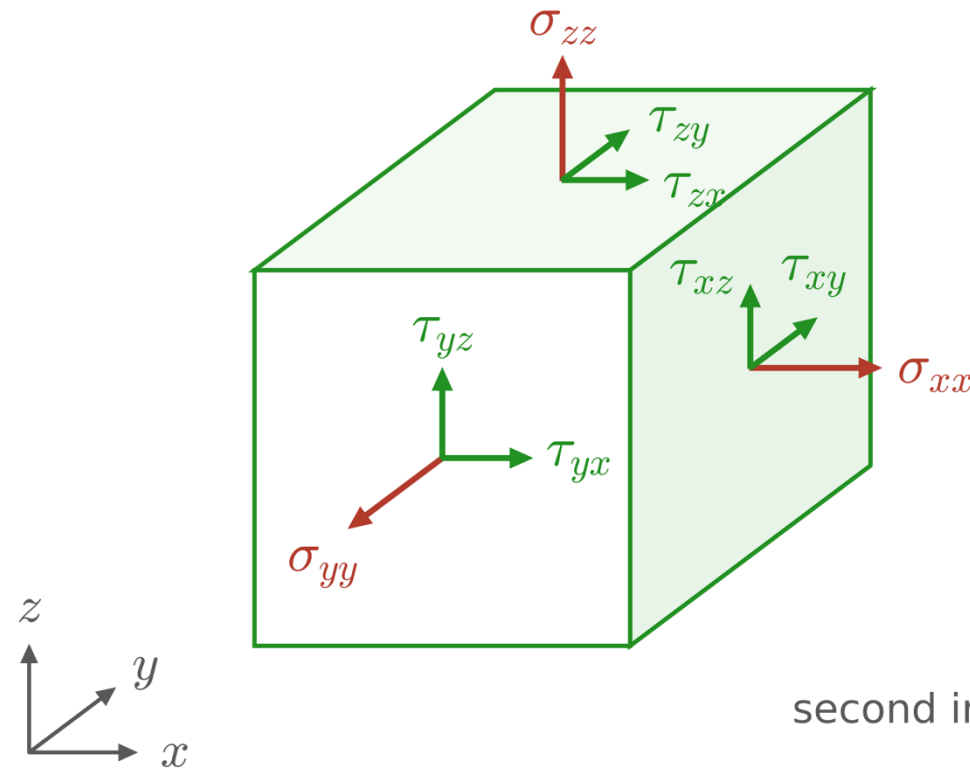
$$\sigma_{ij} = -p \delta_{ij} + \tau_{ij} , \quad \delta_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

First index: the face normal.
Second index: the direction of the force

The diagonal terms are normal stresses;
the rest are shear stresses.

The viscous stress tensor in 3D

Nine components act on the element, three on each of the three faces:



first index: the face
second index: the push direction

red: normal stress

green: shear stress

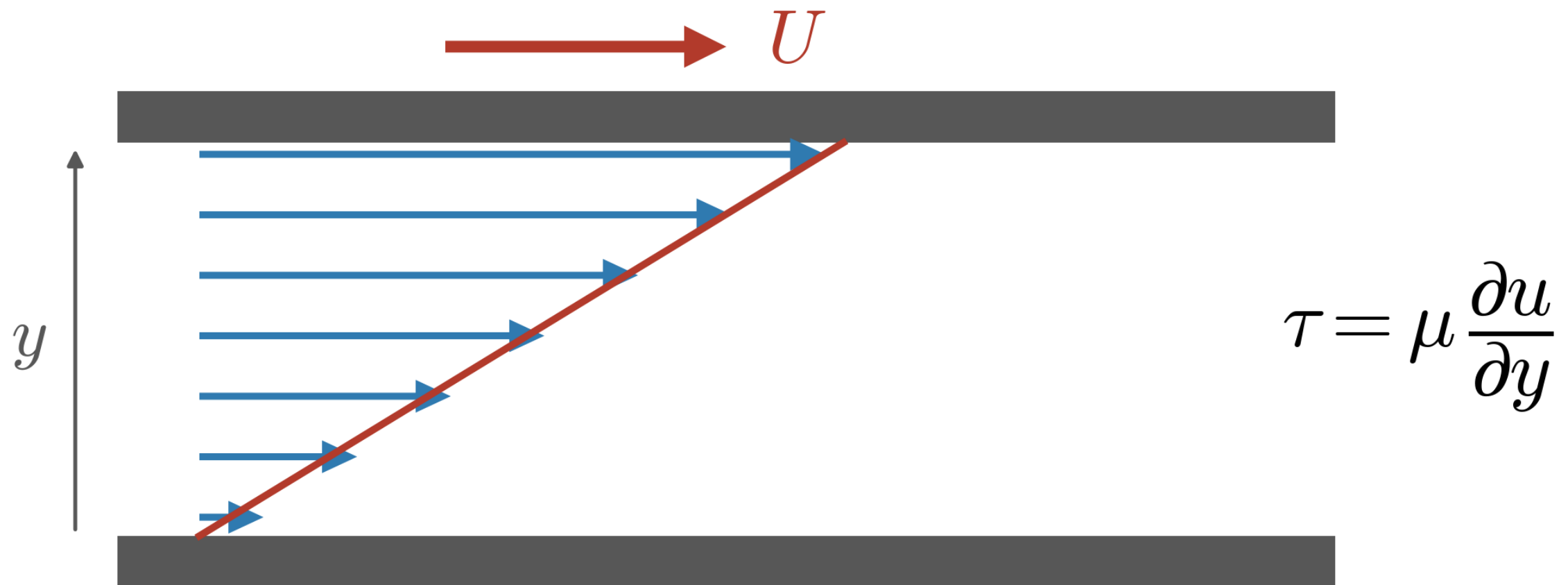
$$\boldsymbol{\sigma} = \begin{bmatrix} \sigma_{xx} & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \sigma_{yy} & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \sigma_{zz} \end{bmatrix} = -p \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} + \begin{bmatrix} \tau_{xx} & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \tau_{yy} & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \tau_{zz} \end{bmatrix}$$

Angular momentum makes the tensor symmetric:
there are 6 independent elements, not 9.

$$\boldsymbol{\sigma} = -p \mathbf{I} + \boldsymbol{\tau} \quad \xrightarrow{\text{Toro}} \quad \mathbf{S} = -p \mathbf{I} + \boldsymbol{\Pi}, \quad \Pi_{ij} \equiv \tau_{ij}$$

The Newtonian fluid

The stress tensor must be related to the motion. For a Newtonian fluid, the viscous stress is proportional to the rate of strain (remember Unit 1).



$$\tau_{ij} = \mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) - \frac{2}{3} \mu \delta_{ij} \frac{\partial u_k}{\partial x_k}, \quad i, j, k = 1, 2, 3$$

μ is the dynamic viscosity. The last term is **Stokes' hypothesis**, which makes the viscous stresses vanish for a uniform expansion.

The 9 components of the viscous stress tensor

For a Newtonian fluid the 3 normal viscous stresses are:

$$\tau_{xx} = 2\mu \frac{\partial u}{\partial x} - \frac{2}{3}\mu \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} \right)$$

$$\tau_{yy} = 2\mu \frac{\partial v}{\partial y} - \frac{2}{3}\mu \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} \right)$$

$$\tau_{zz} = 2\mu \frac{\partial w}{\partial z} - \frac{2}{3}\mu \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} \right)$$

And the 3 independent shear stresses:

$$\tau_{xy} = \tau_{yx} = \mu \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right), \quad \tau_{xz} = \tau_{zx} = \mu \left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x} \right)$$

$$\tau_{yz} = \tau_{zy} = \mu \left(\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y} \right)$$

The Navier–Stokes momentum equation

Written out for the normal and shear components in Cartesian coordinates:

$$\tau_{xx} = \frac{2}{3}\mu \left(2\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y} - \frac{\partial w}{\partial z} \right)$$

$$\tau_{xy} = \mu \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right) = \tau_{yx}$$

Adding the divergence of the viscous stresses to the pressure force gives the momentum equation of a viscous fluid:

$$\boxed{\rho \frac{D\mathbf{v}}{Dt} = -\nabla p + \nabla \cdot \boldsymbol{\tau}}$$

$$\rho \frac{Du}{Dt} = -\frac{\partial p}{\partial x} + \frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z}$$

The incompressible limit

If the density and the viscosity are constant, the divergence term collapses to a Laplacian:

$$\boxed{\rho \frac{D\mathbf{v}}{Dt} = -\nabla p + \mu \nabla^2 \mathbf{v}} \quad \text{with} \quad \nabla \cdot \mathbf{v} = 0$$

In component form, with $\nu = \mu/\rho$ the kinematic viscosity:

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} = -\frac{1}{\rho} \frac{\partial p}{\partial x} + \nu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right)$$

This is the form used for most liquid flows, and for gases below about Mach 0.3.

The compressible Navier–Stokes equations

Same vector equation, same conserved variables.

Only the fluxes change: each one gains the viscous stresses and the heat flux.

$$\Theta_x = \begin{bmatrix} \rho u \\ \rho u^2 + p - \tau_{xx} \\ \rho uv - \tau_{xy} \\ \rho uw - \tau_{xz} \\ (E_t + p)u - u\tau_{xx} - v\tau_{xy} - w\tau_{xz} + q_x \end{bmatrix}$$

The y and z fluxes follow by cycling the indices $x \rightarrow y \rightarrow z$.

The viscous terms contain first derivatives of the velocity and temperature, so the fluxes contain second derivatives of the unknowns.

Adding viscosity turns a first-order hyperbolic system into a second-order one, with a parabolic character in time.

The incompressible Navier–Stokes equations

With constant density, the energy equation decouples and only two equations remain:

$$\nabla \cdot \mathbf{v} = 0$$

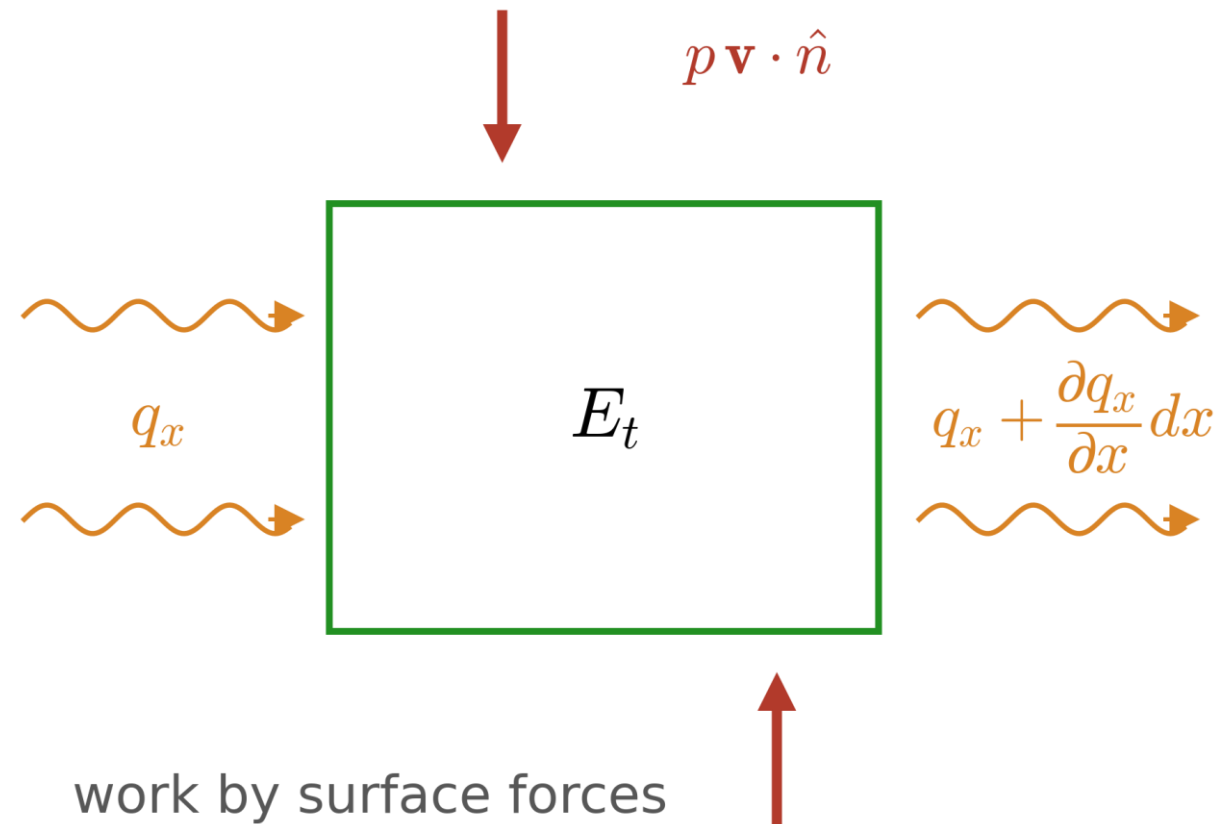
$$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} = -\frac{1}{\rho} \nabla p + \nu \nabla^2 \mathbf{v}$$

There is no evolution equation for the pressure

Taking the divergence of the momentum equation and using the constraint gives one:

$$\nabla^2 p = -\rho \nabla \cdot [(\mathbf{v} \cdot \nabla) \mathbf{v}] = -\rho \frac{\partial u_i}{\partial x_j} \frac{\partial u_j}{\partial x_i}$$

Heat and work



Heat:

Conducted through the faces. Fourier's law relates the flux to the temperature gradient.

$$\mathbf{q} = -k \nabla T$$

$$q_x = -k \frac{\partial T}{\partial x}, \quad q_y = -k \frac{\partial T}{\partial y}, \quad q_z = -k \frac{\partial T}{\partial z}$$

Work:

Done on the faces by pressure and by the viscous stresses, at a rate equal to force times velocity.

Transport coefficients: Sutherland

μ is the dynamic viscosity, k the conductivity in Fourier's law.

Sutherland's law gives both from the temperature alone:

$$\mu(T) = \mu_0 \left(\frac{T}{T_0} \right)^{3/2} \frac{T_0 + S_\mu}{T + S_\mu}$$
$$k(T) = k_0 \left(\frac{T}{T_0} \right)^{3/2} \frac{T_0 + S_k}{T + S_k}$$

*For air: $\mu_0 = 1.716 \times 10^{-5} \text{ Pa s}$, $k_0 = 0.0241 \text{ W m}^{-1} \text{ K}^{-1}$ at $T_0 = 273.15 \text{ K}$;
 $S_\mu = 110.4 \text{ K}$, $S_k = 194 \text{ K}$.*

5 conservation + 2 state + 2 transport relations

9 unknowns: $\rho, u, v, w, e, p, T, \mu, k$

Codes often take k from a constant Prandtl number, $\text{Pr} = \mu c_p / k \approx 0.72$ for air.

In the Euler system $\mu = k = 0$, so neither relation is needed.

The energy equation

Collecting the accumulation, the convection, the conduction and the two kinds of work, all per unit volume:

$$\underbrace{\frac{\partial E_t}{\partial t} + \nabla \cdot (E_t \mathbf{v})}_{\text{rate of change}} = \underbrace{-\nabla \cdot \mathbf{q}}_{\text{heat in}} \underbrace{-\nabla \cdot (p \mathbf{v}) + \nabla \cdot (\boldsymbol{\tau} \cdot \mathbf{v})}_{\text{work by surface forces}}$$

Conservation form again, with everything in one flux:

$$\boxed{\frac{\partial E_t}{\partial t} + \nabla \cdot \left[(E_t + p) \mathbf{v} + \mathbf{q} - \boldsymbol{\tau} \cdot \mathbf{v} \right] = 0}$$

With no conduction and no viscosity only the pressure work survives:

$$\boxed{\frac{\partial E_t}{\partial t} + \nabla \cdot \left[(E_t + p) \mathbf{v} \right] = 0}$$

The internal-energy form

Subtracting the mechanical energy, that is $\mathbf{v}$ dotted into the momentum equation, leaves an equation for the internal energy alone:

$$\rho \frac{De}{Dt} + p \nabla \cdot \mathbf{v} = - \nabla \cdot \mathbf{q} + \Phi$$

Φ is the dissipation function:

the rate at which the viscous stresses convert organised motion into heat.

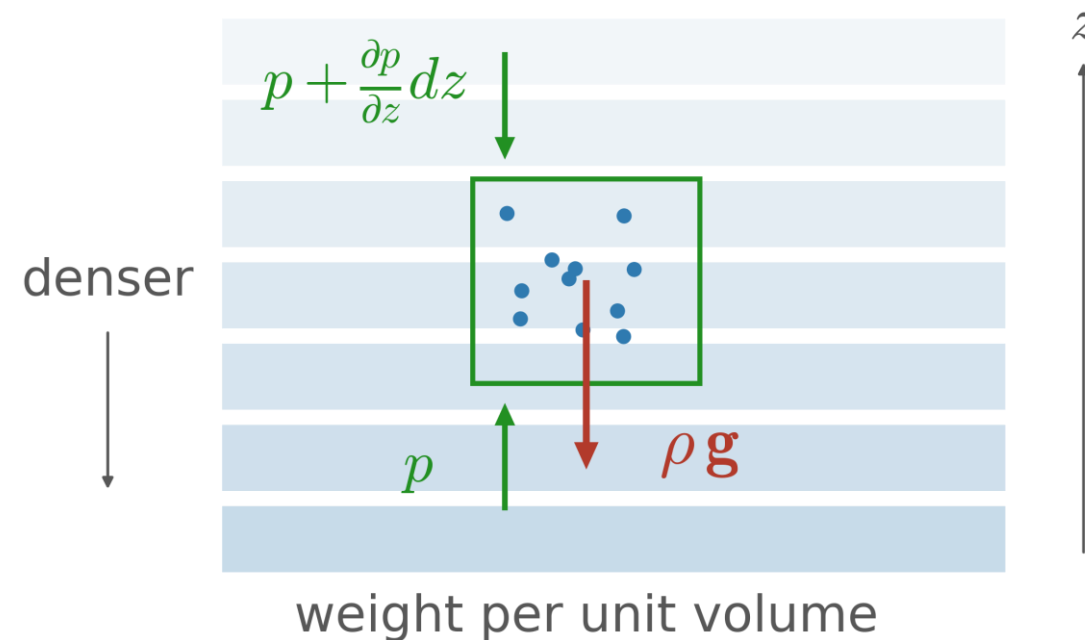
$$\Phi = \tau_{ij} \frac{\partial u_i}{\partial x_j} \geq 0$$

This is never negative.

That is the second law of thermodynamics showing up in the equations of motion.

Body forces: gravity

Everything so far used surface forces. A body force acts on the mass throughout the volume:



$$\rho \frac{D\mathbf{v}}{Dt} = -\nabla p + \nabla \cdot \boldsymbol{\tau} + \rho \mathbf{g}$$

The energy equation gains a source term:

$$\frac{\partial E_t}{\partial t} + \nabla \cdot \left[(E_t + p)\mathbf{v} + \mathbf{q} - \boldsymbol{\tau} \cdot \mathbf{v} \right] = \rho \mathbf{g} \cdot \mathbf{v}$$

At rest this is hydrostatics: $\nabla p = \rho \mathbf{g}$, Lectures 3 and 4.

Body forces: magnetic fields

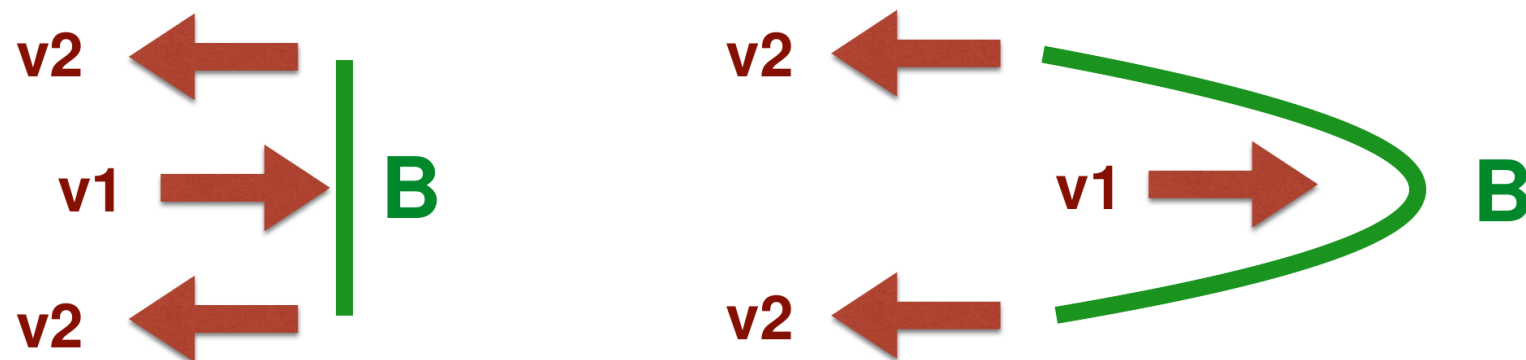
In a conducting fluid the current and the field produce the Lorentz force:

$$\mathbf{f}_L = \mathbf{J} \times \mathbf{B} = \frac{1}{\mu_0} (\nabla \times \mathbf{B}) \times \mathbf{B}$$

$$= \underbrace{\frac{1}{\mu_0} (\mathbf{B} \cdot \nabla) \mathbf{B}}_{\text{magnetic tension}} - \underbrace{\nabla \left(\frac{B^2}{2\mu_0} \right)}_{\text{magnetic pressure}}$$

The field is carried with the fluid; an induction equation is added:

$$\frac{\partial \mathbf{B}}{\partial t} = \nabla \times (\mathbf{v} \times \mathbf{B}) , \quad \nabla \cdot \mathbf{B} = 0$$



Field lines are frozen into the plasma and bend with it.

Summary: Euler and Navier–Stokes

Euler: inviscid, no conduction

Five first-order equations in conservation form.

Hyperbolic in time: finite signal speeds, characteristics, shocks.

No drag, no boundary layer, no dissipation. Slip is allowed at a wall.

Navier–Stokes: viscous, conducting

The same five equations with viscous stresses and heat flux added.

Hyperbolic + Parabolic character in time: second order terms

Drag, boundary layers, separation and turbulence. No slip at a wall.

They represent the same conservation laws.

The difference is entirely in the constitutive relation for the surface stress.

What kind of equations are these?

Unsteady Euler:

Hyperbolic in time. The eigenvalues of the flux Jacobian are u , $u + c_s$ and $u - c_s$.

Steady Euler:

Elliptic where the flow is subsonic, hyperbolic where it is supersonic: the Tricomi problem of Lecture 9.

Unsteady Navier–Stokes:

Mixed. Hyperbolic convection with a parabolic viscous part.

The viscous terms dominate near walls.

Incompressible Navier–Stokes:

Parabolic in the velocity, with an elliptic constraint through the pressure.

The classification decides the numerical method:

We **march** the hyperbolic part, **relax** the elliptic part.

Key equations: Euler

	Compressible	Incompressible
Mass	$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0$	$\nabla \cdot \mathbf{v} = 0$
Momentum	$\frac{\partial(\rho \mathbf{v})}{\partial t} + \nabla \cdot (\rho \mathbf{v} \mathbf{v} + p \mathbf{I}) = 0$	$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} = -\frac{1}{\rho} \nabla p$
Energy	$\frac{\partial E_t}{\partial t} + \nabla \cdot [(E_t + p) \mathbf{v}] = 0$	decoupled ($\rho = \text{const}$)
Closure	$p = (\Gamma - 1) \rho e, \quad c_s = \sqrt{\Gamma p / \rho}$	$\nabla^2 p = -\rho \frac{\partial u_i}{\partial x_j} \frac{\partial u_j}{\partial x_i}$
Type	hyperbolic in time	hyperbolic in $\mathbf{v}$, elliptic in p
Relevance	shocks, supersonic flow, acoustics	low-speed flow, $M < 0.3$

Key equations: Navier–Stokes

	Compressible	Incompressible
Mass	$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0$	$\nabla \cdot \mathbf{v} = 0$
Momentum	$\frac{\partial(\rho \mathbf{v})}{\partial t} + \nabla \cdot (\rho \mathbf{v} \mathbf{v} + p \mathbf{I} - \boldsymbol{\tau}) = 0$	$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} = -\frac{1}{\rho} \nabla p + \nu \nabla^2 \mathbf{v}$
Energy	$\frac{\partial E_t}{\partial t} + \nabla \cdot [(E_t + p) \mathbf{v} + \mathbf{q} - \boldsymbol{\tau} \cdot \mathbf{v}] = 0$	decoupled ($\rho, \nu = \text{const}$)
Closure	$\begin{aligned} \boldsymbol{\tau} &\text{ Newtonian,} & \mathbf{q} &= -k \nabla T \\ p &= (\Gamma - 1) \rho e, & \mu(T), k(T) &\text{ Sutherland} \end{aligned}$	$\begin{aligned} \nabla^2 p &= -\rho \frac{\partial u_i}{\partial x_j} \frac{\partial u_j}{\partial x_i} \\ \nu &= \mu / \rho \end{aligned}$
Type	hyperbolic + parabolic	parabolic in $\mathbf{v}$, elliptic in p
Relevance	boundary layers, drag, heat transfer	liquids, turbulence, $M < 0.3$